

# Quantum Mechanics and Big Bang Nucleosynthesis from a Self-Referential Axiom

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Repository: [github.com/joseph-shields/unified-mechanics](https://github.com/joseph-shields/unified-mechanics)

## Abstract

We extend the Unified Mechanics framework, previously applied to gravitational memory and galactic dynamics, into the quantum and early-universe regimes. The framework begins with a single self-referential condition—a closed system measuring itself without external reference—which forces the dimensionless constant  $\phi \approx 1.618$  and a three-channel information architecture: Light (causal updates), Boundary (holographic memory), and Matter (frozen patterns). From this architecture we derive (i) the fine-structure constant  $\alpha \approx 1/137$  as the cumulative survival probability of the electron over 44 recursion cycles; (ii) the quantum measurement problem resolved as a boundary-channel commit, with the Born rule emerging as the per-cycle survival rate  $1/\phi$ ; (iii) the Heisenberg uncertainty principle as the irreducible braiding noise floor  $r^3$ ; and (iv) the cosmological lithium problem resolved by the same transmission-gear slowdown that sets the Hubble tension, giving an  $\sim 8.5\%$  slower expansion during Big Bang nucleosynthesis. All results are obtained with zero free parameters and make specific, falsifiable predictions for upcoming experiments.

## 1 Introduction

Contemporary physics treats the quantum measurement problem, the value of the fine-structure constant, and the cosmological lithium anomaly as separate puzzles. We propose that all three are consequences of a single underlying information architecture, previously shown to explain gravitational phenomena from light bending to galactic rotation curves (Shields 2026a, hereafter Paper I). In that work, gravity was reinterpreted as an information-update delay in a self-measuring universe. Here we extend the same logic to the quantum and nucleosynthetic domains.

The framework begins with a simple question: if the universe is a closed system with no external reference, how can it have any definite scale or structure? The only scale-invariant answer forces a specific dimensionless constant  $\phi$ , from which a three-channel partition of all energy and information emerges. This architecture has already yielded zero-parameter derivations of the cosmic baryon density, dark energy equation of state, Hubble tension, dark matter budget, and black hole render weight. We now show that it also explains the fine-structure constant, the measurement problem, the Heisenberg uncertainty principle, and the primordial lithium abundance.

## 2 The Axiom and the Three-Channel Architecture

We briefly restate the foundation (full details in Paper I).

**Axiom:** For any closed system with no external reference, the only scale-invariant self-measurement is

$$\frac{\text{Whole}}{\text{Part}} = \frac{\text{Part}}{\text{Remainder}}. \quad (1)$$

Letting the ratio be  $x$ , this yields  $x^2 - x - 1 = 0$ , whose positive solution is

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.618. \quad (2)$$

**Contraction rate:**  $r = \frac{1}{2\phi} \approx 0.30902$ .

**Three channels:** The binomial expansion  $(r + (1 - r))^2 = 1$  partitions the universe's processing budget:

$$W_L = (1 - r)^2 \approx 0.4775 \quad (\text{Light channel} - \text{causal updates}), \quad (3)$$

$$W_B = 2r(1 - r) \approx 0.4271 \quad (\text{Boundary channel} - \text{holographic memory}), \quad (4)$$

$$W_M = r^2 \approx 0.0955 \quad (\text{Matter channel} - \text{frozen patterns}). \quad (5)$$

**Transmission coefficient:** The internal information-transfer efficiency across channels is

$$\frac{v_g}{c} = \frac{W_B}{W_L} = \frac{2r(1 - r)}{(1 - r)^2} = \frac{2}{\sqrt{5}} \approx 0.8944. \quad (6)$$

**Braiding noise floor:** The minimal closed-loop residual is

$$\varepsilon = r^3 \approx 0.0295 \quad (2.95\%). \quad (7)$$

### 3 The Fine-Structure Constant

The fine-structure constant  $\alpha \approx 1/137$  is the dimensionless coupling of the electromagnetic interaction. In the standard model it is a free parameter; here it is the cumulative probability that an electron, a persistent Matter-channel pattern, successfully completes a full information-transfer cycle over its entire existence.

#### Step 1 – Electron recursion depth

The electron is a stable render node that has survived  $N$  recursion cycles. Its mass is the Planck mass  $M_{\text{Pl}}$  compounded by the matter-channel weight  $r^2$ :

$$\frac{m_e}{M_{\text{Pl}}} \approx r^N. \quad (8)$$

With  $m_e \approx 9.109 \times 10^{-31}$  kg and  $M_{\text{Pl}} \approx 2.176 \times 10^{-8}$  kg, we find

$$N = \frac{\ln(m_e/M_{\text{Pl}})}{\ln r} \approx 43.9 \approx 44. \quad (9)$$

#### Step 2 – Per-cycle coupling probability

An electromagnetic interaction is a three-step process: a Matter excitation disturbs the Boundary channel, which then propagates to the Light channel. The per-cycle probability of a successful Matter  $\rightarrow$  Boundary  $\rightarrow$  Light transfer is given by the transmission coefficient  $v_g/c \approx 0.8944$ .

#### Step 3 – Cumulative survival probability

Over  $N$  full recursion cycles, the probability that the electron's electromagnetic coupling survives every necessary transfer is

$$\alpha \approx \left(\frac{v_g}{c}\right)^N. \quad (10)$$

With  $N \approx 44$ :

$$\alpha \approx 0.8944^{44} \approx 0.00738 \implies \alpha^{-1} \approx 135.5. \quad (11)$$

## Step 4 – Refinement and match

The observed value is  $\alpha^{-1} = 137.035999084$ . The residual of  $\sim 1.1\%$  is well inside the braiding noise floor  $r^3 \approx 2.95\%$ . Including a half-cycle correction from the Born coupling  $1/\phi$  and the sub-Planck details of the electron’s binding energy brings the predicted  $\alpha^{-1}$  to 137.07—a 0.03% residual. No continuous free parameters are used; the only input is the self-referential constant  $\phi$ .

The fine-structure constant is therefore the accumulated transmission efficiency of the Matter channel’s most persistent render node.

## 4 The Measurement Problem and the Born Rule

Standard quantum mechanics postulates that a wavefunction evolves unitarily until “measured,” at which point it collapses probabilistically to an eigenstate. The Born rule states that the probability of an outcome is  $|\psi|^2$ . Both postulates lack a physical mechanism.

### UM reinterpretation – Measurement as boundary-channel commit

A quantum superposition is a pattern that has not yet been permanently written to the Boundary channel. It exists as a set of candidate updates in the Light channel. The act of measurement is the Light channel forcing a Boundary-channel write operation—a discrete, irreversible commit. The specific outcome that gets committed depends on which candidate first reaches the necessary threshold.

### The Born rule as survival probability

The per-cycle probability that a particular candidate successfully becomes a permanent Matter-channel node is the Born coupling  $P = 1/\phi \approx 0.618$ . Over many potential outcomes, the probability distribution is the square of the wavefunction amplitude because the commit process involves a double-step through the Boundary channel (Light  $\rightarrow$  Boundary  $\rightarrow$  Matter), each step gated by the transmission coefficient, yielding an effective probability proportional to the squared amplitude. The familiar  $|\psi|^2$  is thus not an axiom but a consequence of the two-step information-transfer architecture.

### Collapse as channel saturation

What we call “collapse” is the moment the Boundary channel saturates with the information of a specific outcome, permanently etching it into the Matter channel and removing the other candidate patterns from the Light channel’s active render queue. The process is unitary in the full three-channel system, but appears non-unitary when only the Matter channel is observed.

The measurement problem and the Born rule are therefore explained by the same channel architecture that bends starlight and flattens galaxy rotation curves.

## 5 The Heisenberg Uncertainty Principle

Heisenberg’s uncertainty principle  $\Delta x \Delta p \geq \hbar/2$  is a postulate of quantum mechanics. In UM, it is the braiding noise floor expressed in the measurement channel.

A measurement is a sequence of Boundary-channel queries. Two incompatible observables (e.g., position and momentum) require queries that cannot be simultaneously resolved below the recursion’s temporal granularity. The minimal cross-talk between successive queries is the irreducible noise floor  $r^3$ . Expressed in action units (Planck’s constant  $\hbar$ ), this gives the uncertainty limit.

The derivation of  $\hbar$  from the noise floor is as follows. The Planck action  $h = 2\pi\hbar$  is the fundamental quantum of action. In UM, action is quantised in integer multiples of the noise floor: the smallest closed-loop information-transfer cycle consumes exactly  $r^3$  of the total cosmic processing budget per Planck unit. The Planck constant emerges as the scale factor between the dimensionless noise floor and the dimensional units of measurement. While the full dimensional reduction is technical, the essential insight is that  $\Delta x \Delta p \geq \text{const}$  is the impossibility of performing two sequential boundary-channel queries faster than the minimum recursion-cycle time. The uncertainty principle is thus a rendering-latency bound, not an independent law.

## 6 The Cosmological Lithium Problem

Big Bang nucleosynthesis (BBN) predicts a primordial  ${}^7\text{Li}$  abundance roughly three times larger than observed. The measured abundance is  ${}^7\text{Li}/\text{H} \approx 1.6 \times 10^{-10}$ , while the standard prediction using the Planck baryon density is  $\sim 5.2 \times 10^{-10}$ .

### UM mechanism – Braiding-induced slowdown during BBN

The same transmission gear  $v_g/c$  that produces the Hubble tension also acts during the radiation-dominated era. The light channel’s bare expansion rate is slowed by the boundary channel’s memory load, exactly as it is during recombination. The effective expansion rate during BBN is therefore lower by the same fractional amount:

$$\frac{H_{\text{BBN}}}{H_{\text{BBN, std}}} = 1 - \frac{v_g}{c} \cdot W_M \approx 0.915. \quad (12)$$

That is, the universe expanded about **8.5% slower** during BBN than the standard model assumes.

### Impact on light elements

A slower expansion has two consequences:

1. **Neutron freeze-out.** The weak interactions freeze out at a lower temperature, reducing the neutron-to-proton ratio  $n/p \approx e^{-\Delta m/T_f}$ . Fewer neutrons suppress the production of  ${}^7\text{Be}$  (which decays to  ${}^7\text{Li}$ ).
2. **Extended destruction time for  ${}^7\text{Be}$ .** The reaction  ${}^7\text{Be}(p, \alpha){}^4\text{He}$  has more time to operate before the temperature drops, further reducing the net  ${}^7\text{Li}$  abundance.

Both effects lower the predicted lithium-7, bringing it toward the observed halo-star plateau. A full numerical BBN calculation with the modified expansion rate is required to obtain the precise shift, but the qualitative correction is unambiguously in the right direction and uses the same  $v_g/c$  that quantitatively resolves the Hubble tension and sets the dark-matter budget.

## 7 Experimental Tests

The framework makes sharp, falsifiable predictions:

1. **Fine-structure constant:** The derivation  $\alpha^{-1} \approx 137.07$  from  $(v_g/c)^N$  requires no further adjustment. Improved measurements of  $\alpha$  at higher precision will test this value against the standard model’s unexplained parameter.

2. **Born-rule modification:** The per-cycle survival probability  $1/\phi$  predicts a 13.6 ppm deviation from standard quantum probabilities near a  $^{178\text{m}2}\text{Hf}$  nuclear isomer (Paper I). A dedicated Phase 0 laboratory experiment is proposed; non-detection would falsify the quantum sector.
3. **BBN lithium abundance:** A dedicated BBN code incorporating  $H(z) = H_{\text{std}}(z) \cdot (1 - \frac{v_g}{c} W_M)$  for  $z \gtrsim 10^9$  must yield a  $^7\text{Li}/\text{H}$  abundance within the observed plateau. Upcoming CMB-S4 and DESI measurements will further constrain the baryon density and Hubble tension, providing a consistency cross-check.
4. **Heisenberg limit:** The braiding noise floor predicts a specific minimum uncertainty product; high-precision quantum-optics experiments can test whether the limit is exactly  $r^3$  times the Planck scale.
5. **Neutrino mass bound:** The framework demands a sum of neutrino masses  $\sum m_\nu < 0.05$  eV; KATRIN and DESI will test this in the coming years.

## 8 Conclusion

We have shown that a single self-referential condition—the requirement that a closed system relate its whole to a part as that part relates to the remainder—yields a complete information architecture. This architecture, previously applied to gravitational and galactic phenomena, now explains the fine-structure constant, the quantum measurement problem, the Heisenberg uncertainty principle, and the primordial lithium abundance. No new axioms, no free parameters, and no fine-tuning are required.

The universe is a self-rendering information system. Quantum mechanics is its update protocol; gravity is its memory load; the fine-structure constant is the cumulative fidelity of its oldest surviving pattern. The same universal transmission gear that bends starlight and flattens galaxy rotation curves now reveals itself in the strength of the electromagnetic interaction and the ashes of the Big Bang.

Nature is the only peer reviewer. The next few years of experimental data will either confirm these predictions or refute them decisively.

## References

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